Shubham Innani

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Professional Summary

AI-in-Healthcare leader with 5+ years developing deep learning models into tools. Led development of a whole-slide image (WSI) diagnostic pipeline trained on **2,500+ multi-site patient cases**; delivered a clinician-facing user interface and GPU-backed inference pipeline (A100/H100/A6000 SLURM clusters) that performs inference in minutes, reducing diagnostic turnaround from weeks to minutes and progressing toward Indiana University Health clinical validation.

Education

2016-2020 **Bachelor of Technology**, *Department of Electronics and Telecommunication Engineering*, *Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded*, Maharashtra, India.

Experience

December 2023 - Present **Research Data Analyst**, *Division of Computational Pathology, Indiana University School of Medicine*, Indianapolis, United States of America.

- Led development and translation of interpretable AI models for whole-slide images into a deployable clinical research tool supporting WHO-aligned diagnostic classification and prognostic stratification; trained and validated models on 2,500+ brain-tumor cases aggregated from >25 clinical sites.
- Built a clinician-facing UI and GPU-backed inference pipeline (A6000 / A100 on Quartz and BigRed200 SLURM clusters) enabling WSI inference in 10 minutes per slide, compared to multi-week turnaround for standard molecular workflows; demonstrated to 3 clinicians and staged for IU Health clinical validation. **News:** `https://medicine.iu.edu/news/2025/10/bakas-ai-glioma-identification`
- Translated complex model outputs into actionable clinical insights for pathologists and research PIs, supporting peer-reviewed publications and evidence generation toward clinical utility.
- Implemented reproducible, HPC-friendly training and inference workflows with standardized documentation, enabling scalable reuse across projects and institutions.

August 2022 - November 2023 **Visiting Associate**, *Center for Biomedical Image Computing and Analytics (CBICA), University of Pennsylvania*, Philadelphia, United States of America.

- Developed multimodal prognostic models for glioblastoma (imaging + clinical features), achieving an AUC ≈ 0.75 on test data (188 GBM patients across multi-institutional cohorts); co-authored manuscripts in Modern Pathology and Frontiers.
- Led imaging-side implementation, coordinated model experiments with collaborators, and delivered code for downstream validation.

January 2021- July 2022 **Assistant System Engineer**, *Tata Consultancy Services*, Pune, India.

- QA lead for cloud migration of transaction modules (legacy → AWS, Java/Spring Boot) for a leading U.S. financial client; owned API and end-to-end transaction verification, ensuring parity between legacy and cloud systems.
- Led a 6-person testing component across a multi-team cloud migration, improving deployment reliability and reducing post-migration incidents through comprehensive API and transaction-level validation.
- Collaborated with cross-functional teams and business Subject Matter Experts to validate complex transaction flows and support compliance requirements during migration.

March 2020- June 2020 **Data Science Intern**, *Nymo.ai*, Bangalore, India.

- Built edge inference pipelines for Automatic Number Plate Recognition (ANPR) and mask-detection; supported prototype-to-deployment deliveries for clients (one deployment delivered in ∼2 weeks).
- Delivered real-time inference (approx. 6-8 fps on edge devices), and implemented OCR integration for plate-to-text extraction (model accuracy ∼73% during internship).

Awards

2025 Best Paper Award for "AI-driven WHO 2021 classification of gliomas based only on H&E-stained slides" at Center for Computational Biology and Bioinformatics Retreat (CCBB), Indiana University, Indianapolis

2025 **5th prize** in LLM based report generation in pathology challenge (REG) held at MICCAI 2025, the 28th International Conference on Medical Image Computing and Computer Assisted Intervention.

2020 **2nd Prize** in semantic segmentation challenge on IDD dataset organized at Indian Conference on Computer Vision, Graphics and Image Processing (ICVGIP) at IIT Jodhpur, by IIIT Hyderabad, and Intel, India.

2020 Ranked among top 10 in retinal fundus glaucoma challenge (REFUGE2) held at MICCAI 2025, the 23rd International Conference on Medical Image Computing and Computer Assisted Intervention.

2020 Ranked among 10 in age related macular degeneration (AMD) challenge held at IEEE ISBI 2020, the International Symposium on Biomedical Imaging.

2019 **1st Prize** in semantic segmentation challenge on IDD lite dataset organized at National Conference on Computer Vision, Pattern Recognition, Image Processing and Graphics (NCVPRIPG),India, hosted by IIIT Hyderabad and Intel.

2019 Travel Grant awarded to attend National Conference on Computer Vision, Pattern Recognition, Image Processing and Graphics (NCVPRIPG).

Publications

Journal

2026 **Shubham Innani**, MacLean Nasrallah, W. Robert Bell, Bhakti Baheti*, Spyridon Bakas*, AI-driven WHO 2021 classification of gliomas based only on H&E-stained slides, in Neuro-Oncology, Volume 28, Issue 1, January 2026, Pages 282-296

2025 **Shubham Innani**, MacLean Nasrallah, W. Robert Bell, Bhakti Baheti*, Spyridon Bakas*, Interpretable artificial intelligence based determination of glioma IDH mutation status directly from histology slides, in Neuro-Oncology Advances 7.1 (2025): vdaf140.

2025 Bhakti, Baheti, Sunny Rai, **Shubham Innani**, Garv Mehdiratta, Sharath Chandra Guntuku, MacLean P. Nasrallah, and Spyridon Bakas, Multimodal Explainable AI for Prognostic Stratification of Glioblastoma Patients, In Modern Pathology (2025): 100797

2025 Katrina Collins*, **Shubham Innani***, Kingsley Ebare, Mohammed Saad, Stephanie E. Siegmund, Sean R. Williamson, Fiona Maclean, Andres Matoso, Ankur Sangoi, Michelle S. Hirsch, Dibson D. Gondim, Andres M. Acosta, Bhakti Baheti, Spyridon Bakas, Muhammad T. Idrees; Artificial Intelligence–Based Classification of Renal Oncocytic Neoplasms: Advancing From a 2-Class Model of Renal Oncocytoma and Low-Grade Oncocytic Tumor to a 3-Class Model Including Chromophobe Renal Cell Carcinoma. Arch Pathol Lab Med 2025; doi: https://doi.org/10.5858/arpa.2024-0374-OA(*contributed equally)

2024 Bhakti Baheti, **Shubham, Innani**, MacLean P. Nasrallah, and Spyridon Bakas, Prognostic stratification of glioblastoma patients by unsupervised clustering of morphology patterns on whole slide images furthering our disease understanding, In Frontiers in Neuroscience 18 (2024): 1304191.

2023 **Shubham, Innani**, Prasad Dutande, Ujjwal Baid, Venu Pokuri, Spyridon Bakas, Sanjay Talbar, Bhakti Baheti, and Sharath Chandra Guntuku, Generative adversarial networks based skin lesion segmentation In Nature Scientific Reports 13, no. 1 (2023): 13467.

2022 Fang H, Li F, Fu H, Sun X, Cao X, Lin F, Son J, Kim S, Quellec G, Matta S, Shankaranarayana SM, Ujjwal Baid, **Shubham Innani** *et al.*, ADAM Challenge: Detecting Age-related Macular Degeneration from Fundus Images, In IEEE Transactions on Medical Imaging, doi: 10.1109/TMI.2022.3172773.

2020 Bhakti Baheti, **Shubham Innani**, Suhas Gajre, Sanjay Talbar, Semantic scene segmentation in unstructured environment with modified DeepLabV3+, In Elsevier Pattern Recognition Letters, Volume 138, 2020, Pages 223-229, ISSN 0167-8655,doi: 10.1016/j.patrec.2020.07.029.

Conference

2024 **Shubham Innani**, MacLean Nasrallah, W. Robert Bell, Bhakti Baheti*, Spyridon Bakas*, Multi-scale Whole Slide Image Assessment Improves Deep Learning based WHO 2021 Glioma Classification, In MICCAI Workshop on Computational Pathology with Multimodal Data (COMPAYL), 2024

2024 **Shubham Innani**,Bhakti Baheti, MacLean Nasrallah, Spyridon Bakas, Weakly supervised IDH-status glioma classification from H&E-stained whole slide images, In 21st IEEE International Symposium on Biomedical Imaging (ISBI 2024), Athens, Greece

2022 **Shubham Innani**, Prasad Dutande, Bhakti Baheti, Ujjwal Baid and Sanjay Talbar, Deep Learning based Novel Cascaded Approach for Skin Lesion Analysis, In 7th International Conference on Computer Vision & Image Processing, Nagpur, India, 2022

2021 Bhakti Baheti, **Shubham Innani**, Suhas Gajre and Sanjay Talbar, Pedestrian Detection and Movement Direction Recognition with Convolutional Neural Network In 9th International Conference on Pattern Recognition and Machine Intelligence, 2021 December

2021 **Shubham Innani**, Prasad Dutande, Bhakti Baheti, Sanjay Talbar, and Ujjwal Baid, Fuse-PN: A Novel Architecture for Anomaly Pattern Segmentation in Aerial Agricultural Images, In Proceedings of the IEEE CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops(pp. 2960-2968)

2020 M. T. Chiu,X. Xu, K. Wang, J.Hobbs, Brunner R., Andrew Ng, Ujjwal Baid, **Shubham Innani**, Prasad Dutande, Bhakti Baheti, Sanjay Talbar, The 1st Agriculture-Vision Challenge: Methods and Results, In 2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), 2020, pp. 212-218, doi: 10.1109/CVPRW50498.2020.00032.

2020 Bhakti Baheti*, **Shubham Innani***, Suhas Gajre and Sanjay Talbar, Eff-UNet: A Novel Architecture for Semantic Segmentation in Unstructured Environment, In 2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), 2020, pp. 1473-1481, doi: 10.1109/CVPRW50498.2020.00187.(*contributed equally)

Abstract

2025 **Shubham Innani**, W. Robert Bell, MacLean Nasrallah, Bhakti Baheti, Spyridon Bakas; Abstract 6247: Artificial intelligence predicts 2021 WHO glioma subtypes from whole slide images. Cancer Res 15 April 2025; 85 (8_Supplement_1): 6247. https://doi.org/10.1158/1538-7445.AM2025-6247

2024 **Shubham Innani**, Bhakti Baheti, MacLean P Nasrallah, W Robert Bell, Spyridon Bakas, PATH-39. AI-based identification of glioma IDH mutational status from H&E-stained whole slide images, Neuro-Oncology, Volume 26, Issue Supplement_8, November 2024, Page viii187, https://doi.org/10.1093/neuonc/noae165.0738

2024 Siddhesh Thakur, Shahriar Faghani, Mana Moassefi, Ujjwal Baid, Verena Chung, Sarthak Pati, **Shubham Innani**, Bhakti Baheti, Jacob Albrecht, Alexandros Karargyris, Hasan Kassem, Maclean Nasrallah, Jared Ahrendsen, Valeria Barresi, Maria Gubbiotti, Giselle Lopez, Calixto-Hope Lucas, Michael Miller, Lee Cooper, Jason Huse, William Bell, Spyridon Bakas, TMIC-60. Brats-Path: assessing hetegenenous histopathologic regions in glioblastoma, Neuro-Oncology, Volume 26, Issue Supplement_8, November 2024, Page viii312, https://doi.org/10.1093/neuonc/noae165.1238

2023 **Shubham Innani**, Bhakti Baheti, MacLean P Nasrallah, Spyridon Bakas, Interpretable IDH classification from H&E-stained histology slides, Neuro-Oncology, Volume 25, Issue Supplement_5, November 2023, Page v177, https://doi.org/10.1093/neuonc/noad179.0669

2023 Bhakti Baheti, Sunny Rai, **Shubham Innani**, Garv Mehdiratta, Sharath Chandra Guntuku, MacLean P Nasrallah, Spyridon Bakas, Detecting Histologic & Clinical Glioblastoma Patterns of Prognostic Relevance, Neuro-Oncology, Volume 25, Issue Supplement_5, November 2023, Page v126, https://doi.org/10.1093/neuonc/noad179.0478

2023 Bhakti Baheti, **Shubham Innani**, Garv Mehdiratta, MacLean P. Nasrallah, Spyridon Bakas, Interpretable whole slide image prognostic stratification of glioblastoma patients furthering disease understanding, Neuro-Oncology, Volume 25, Issue Supplement_2, September 2023, Pages ii103–ii104 https://doi.org/10.1093/neuonc/noad137.347.

2023 Bhakti Baheti, **Shubham Innani**, MacLean P. Nasrallah, Spyridon Bakas, Unsupervised clustering of morphology patterns on whole slide images guide prognostic stratification of glioblastoma patients, Neuro-Oncology, Volume 25, Issue Supplement_2, September 2023, Page ii15, https://doi.org/10.1093/neuonc/noad137.043.

Arxiv

2024 Spyridon Bakas, Siddhesh P. Thakur, Shahriar Faghani, Mana Moassefi, Ujjwal Baid, Verena Chung, Sarthak Pati, **Shubham Innani**, Bhakti Baheti et al, BraTS-Path Challenge: Assessing Heterogeneous Histopathologic Brain Tumor Sub-regions, arXiv preprint arXiv:2405.10871 (2024).

2024 Bran, Hongwei, Fernando Navarro, Ivan Ezhov, Amirhossein Bayat, Dhritiman Das, Florian Kofler, Suprosanna Shit et al, QUBIQ: Uncertainty Quantification for Biomedical Image Segmentation Challenge, arXiv preprint arXiv:2405.18435 (2024).

2022 Fang H, Li F, Fu H, Sun X, Cao X, Son J, Yu S, Zhang M, Yuan C, Bian C, Lei B., Ujjwal B, **Shubham Innani** and others, REFUGE2 Challenge: Treasure for Multi-Domain Learning in Glaucoma Assessment. arXiv preprint arXiv:2202.08994. 2022 Feb 18.

Presentations

2025 Oral & Poster Presentation at 30th annual meeting of the Society for Neuro-Oncology (SNO 2025), Honolulu, Hawai

2025 Oral Presentation at 21st European Congress on Digital Pathology (ECDP 2025), Barcelona, Spain

2025 Poster Presentation at 116th annual meeting of American Association for Cancer Research (AACR), Chicago, Illinios

2024 Oral Presentation at 29th annual meeting of the Society for Neuro-Oncology (SNO 2024), Houston, Texas

2024 Two Oral Presentation at 20th European Congress on Digital Pathology (ECDP 2024), Vilnius, Lithuania.

2023 Oral Presentation & Poster at 19th European Congress on Digital Pathology (ECDP 2023), Budapest, Hungary.

Projects

- **Vision Language Model based report generation (REG 2025 challenge)**
  Vision language model developed to generate reports from gigapixel whole slide images.
- **Prognostic Risk Stratification of Gliomas**
  AI based prognostic risk stratification in adult-type diffuse gliomas with various pathology foundation models and multiple instance learning models.
- **Multiscale assessment of Whole Slide Image**
  Multiscale assessment of Whole Slide Image to identify WHO 2021 classification of glioma patients with various pathology foundation models and multiple instance learning models.

- **Oncocytic Renal Tumor Subtyping**
  Focus is to develop a computational algorithm for subtyping oncocytic renal tumors from H&E stained whole slide images (WSI).
- **Biomarker Identification in Gliomas**
  Focus is to develop a computational algorithm for biomarker prediction of glioma patients based on digital H&E stained whole slide images (WSI) using weakly supervised approach. We explore Self Supervised Learning based algorithms based on ViT and DINO.
- **Prognostic Stratification of Glioblastoma**
  Focus is to develop a computational algorithm for survival prediction of GBM patients based on digital H&E stained whole slide images (WSI) using weakly supervised approach. We also aim at interpretability analysis i.e. the important regions from WSI driving the decision.
- **Automatic Number Plate Recognition**
  Focus was on the development of novel and realtime algorithms for Automatic Number Plate Recognition. This algorithm implements end-to-end You Only Look Once (YOLO) based object detector and Mask-RCNN based Region Proposal Network.
- **Retinal Fundus Glaucoma Analysis (REFUGE2 Challenge, MICCAI 2020)**
  Algorithm for Retinal Fundus Glaucoma Analysis with segmentation of optic disc and cup, classification of glaucoma and localization of fovea is being developed. We leverage the power of Efficientnets and Unet for all the tasks.
- **Anamoly Pattern Segmentation in Agriculture(Agriculture Vision Challenge, CVPRW 2020)**
  Novel and effective algorithms is developed for agricultural pattern recognition from aerial images using Feature Pyramid Network as encoder and varying the decoder backbones.
- **Age Related Macular Degeneration (AMD Challenge, ISBI 2020)**
  Development of algorithms associated with the diagnosis of Age-related Macular degeneration (AMD) and segmentation of lesions in fundus photos from AMD patients with Convolutional Neural Networks.
- **Semantic Scene Segmentation in Unstructured Environment (AUTONUE Challenge, ICCV 2019 & NCVPRIPG Challenge 2019 & ICVGIP Challenge 2020)**
  Encoder-Decoder architecture is developed for semantic scene segmentation in unstructured environment with various backbones like InceptionResNetV2, EfficientNets using Deep Learning on Indian Driving Dataset (IDD) and IDD lite.
- **Skin lesion Segmentation and Classification (ISIC Challenge, MICCAI 2018)**
  Generative Adversarial Network based architecture is developed for skin lesion segmentation and classification with Convolutional Neural Network. The classification task is carried out with state-of-the-art networks like ResNets, MobileNets, Xception, EfficientNets.

Community Engagements

Reviewer	IEEE Transaction in Medical Imaging, Nature Scientific Reports, IEEE Signal Processing Letters, IEEE Transactions on Neural Networks and Learning Systems, IEEE Computer Vision and Pattern Recognition (Workshops) Conference, IEEE International Symposium on Biomedical Imaging Conference, Elsevier Neurocomputing Journal, Medical Image Computing and Computer Assisted Intervention(MICCAI), npj Precision Oncology, BMC Cancer
Program Committee	2nd International Workshop on Medical Optical Imaging and Virtual Microscopy Image Analysis (MOVI) @ MICCAI 2024
Volunteer	NCI Informatics Technology for Cancer Research (ITCR) annual meeting 2024
Member	MICCAI Special Interest Group on Computational Pathology, Society of Neuro-Oncology (2025-2026), European Congress for Digital Pathology (2024-2026), American Association of Cancer Research (2025), MICCAI Medical Image Computing and Computer Assisted Intervention Society (2024), IEEE Signal Processing Society (2024)

Personal Details

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Google Scholar	`https://scholar.google.com/citations?user=Aw0CjT8AAAAJ&hl=en`
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